

Name: _____

Date: _____

PERCENTAGES, (PERCENTAGE INCREASE AND DECREASE)

LO: I can increase and decrease amounts by a given percentage using multipliers.

Percentage change

To **increase** by a percentage, multiply by **1 + (percentage as decimal)**. For example, increase by 20% means multiply by 1.20.

To **decrease** by a percentage, multiply by **1 - (percentage as decimal)**. For example, decrease by 15% means multiply by 0.85.

Quick examples

●	£80 increased by 25% = $80 \times 1.25 =$ £100	multiply by 1.25
●	£60 decreased by 10% = $60 \times 0.9 =$ £54	multiply by 0.90
●	VAT at 20%: $£50 \times 1.20 =$ £60	add 20%
●	$£80 + 25\% =$ £105	25% of 80 is 20, not 25

Key idea

Use a single multiplier for efficiency. Increase: multiplier > 1.
Decrease: multiplier < 1.

$$\text{New amount} = \text{Original} \times \text{multiplier}$$

e.g. +30% $\times 1.30$, -30% $\times 0.70$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - percentage increase

Increase £450 by 12%.

$$\begin{aligned}\text{Multiplier} &= 1 + 0.12 = 1.12 \\ 450 \times 1.12 \\ &= £504\end{aligned}$$

£504

Add the percentage rate as a decimal to 1.

Example 2 - percentage decrease

A £85 jacket has 20% off in a sale.
Find the sale price.

$$\begin{aligned}\text{Multiplier} &= 1 - 0.20 = 0.80 \\ 85 \times 0.80 \\ &= £68\end{aligned}$$

£68

Subtract the decimal from 1 for a decrease.

Example 3 - finding percentage change

A house was bought for £200,000 and sold for £230,000. Find the percentage increase.

$$\begin{aligned}\text{Change} &= 230,000 - 200,000 = \\ &30,000\end{aligned}$$

$$\begin{aligned}\text{Percentage} &= (30,000 \div 200,000) \times \\ &100 \\ &= 15\%\end{aligned}$$

15% increase

Divide the change by the original, then multiply by 100.

Example 4 - reverse percentage

After a 20% increase, a price is £96.
Find the original price.

$$\begin{aligned}£96 &= \text{original} \times 1.20 \\ \text{Original} &= 96 \div 1.20 \\ &= £80\end{aligned}$$

£80

Divide by the multiplier to reverse a percentage change.

YOUR TURN – PRACTICE (deliberate practice)

1. Percentage increase

Increase each amount by the given percentage.

- a) £200 by 15%
- b) £350 by 8%
- c) £120 by 25%
- d) £600 by 5%

2. Percentage decrease

Decrease each amount by the given percentage.

- a) £500 by 10%
- b) £80 by 25%
- c) £240 by 15%
- d) £1200 by 30%

3. Finding percentage change

Find the percentage change (state increase or decrease).

- a) From £40 to £50
- b) From £80 to £60
- c) From £120 to £150
- d) From £200 to £170

4. Reverse percentages

Find the original amount.

- a) After 10% increase, price is £55
- b) After 20% decrease, price is £64
- c) After 25% increase, price is £75
- d) After 15% decrease, price is £510

5. Real-world problems

Solve each problem.

- a) A £45 shirt has 20% off. Find sale price.
- b) A meal costs £35 plus 12.5% service. Find total.
- c) A population of 50,000 grows by 3%. New population?
- d) After a 5% pay rise, salary is £31,500. Original?
- e) House value increases 8% to £270,000. Original?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To increase by 15%, multiply by 1.15 ☐

Correct answer: _____

Reason: _____

b) Decreasing by 20% then increasing by 20% returns to the original ☐

Correct answer: _____

Reason: _____

c) Percentage change = (change ÷ original) × 100 ☐

Correct answer: _____

Reason: _____

d) To decrease by 30%, multiply by 1.30 ☐

Correct answer: _____

Reason: _____

e) A reverse percentage problem requires dividing by the multiplier ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A shop has a sale with **30% off** everything. A customer also has a **10% loyalty discount** applied after the sale price. If the original price is **£200**, what is the final price? Is this the same as 40% off?

Expression: _____

Simplified: _____

b) The population of a town was **45,000**. It increased by **12%** one year and then decreased by **8%** the next. Find the population after 2 years. Has the overall population increased or decreased?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

